Report

Project: Solar Power Generation System Using MATLAB/Simulink

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Software: MATLAB/Simulink [R2024a]

Abstract

This project presents a solar photovoltaic (PV) power-generation system developed in MATLAB/Simulink R2024a. The model receives irradiance and temperature inputs at the PV array. The generated DC power is supplied to a DC–DC boost-converter stage containing an inductor, controlled semiconductor switch, diode, and DC-link capacitor. An inverter converts the DC-link power into AC, which is supplied to a load. The output voltage waveform is examined both without an output passive filter and with a passive filter. The simulation plots show the pre-boost voltage, post-boost voltage, and inverter output voltage. The filtered output appears more sinusoidal than the unfiltered switching waveform. Numerical performance values such as efficiency, ripple, and RMS voltage should be added only after they are measured from the simulation.

1. Objectives :

- Build and document a PV power-conversion model.
- Study PV response to changes in irradiance and temperature.
- Examine boost-converter and DC-link behaviour.
- Observe inverter output and filtering performance.
- Compare input and output power using consistent measurement intervals.
- Validate the model with repeatable test cases.

2. System description

The Simulink model contains a PV Array block with irradiance (Ir) and temperature (T) inputs. The PV output is connected to a DC–DC boost-converter stage made up of an inductor, a controlled semiconductor switch, a diode, and a DC-link capacitor. The DC link supplies an inverter/bridge, whose output is connected to a load. Two configurations are considered: one without a passive filter at the output and one with a passive filter formed by an inductor and capacitor. Voltage measurement blocks and Scope blocks are used to observe the voltage before boosting, after boosting, and at the output. The exact inverter configuration and control method should be stated from the actual block parameters.

2. Theory :

3.1 Photovoltaic generation

A PV cell converts incident solar radiation into electrical energy. Irradiance strongly affects the generated current and available power. For a typical silicon PV module, increasing cell temperature tends to reduce voltage and maximum power, although exact behaviour depends on the selected PV model.

3.2 Boost converter

For an ideal continuous-conduction boost converter, the approximate relationship is:

`Vout = Vin / (1 − D)`, where `D` is the switch duty ratio and `0 ≤ D < 1`. This equation is an ideal steady-state approximation; parasitic losses, discontinuous conduction, and control dynamics cause deviations.

3.3 DC-link capacitor

The DC-link capacitor stores energy and helps reduce voltage variation between the converter and inverter. Its voltage ripple depends on capacitance, power balance, switching/control behaviour, and load.

3.4 Inverter and filter

The inverter converts DC power into AC using semiconductor switching. PWM, if implemented, controls the output waveform. The filter attenuates switching-frequency components. Report the actual topology and control method present in the model.

4. Model implementation :

The model was implemented in MATLAB/Simulink R2024a. The following implementation details should be recorded from the actual block masks before submission:

- Software: MATLAB/Simulink R2024a. Solver type, step size/tolerances, and simulation stop time: [enter actual settings].

- PV Array: the supplied model screenshot shows irradiance of 1500 W/m² and temperature of 25°C. Record the PV module/model and array series/parallel configuration from the block.

- Boost converter: record the inductor and capacitor values, switching frequency, pulse duty ratio, and whether the control is open-loop or feedback-based.

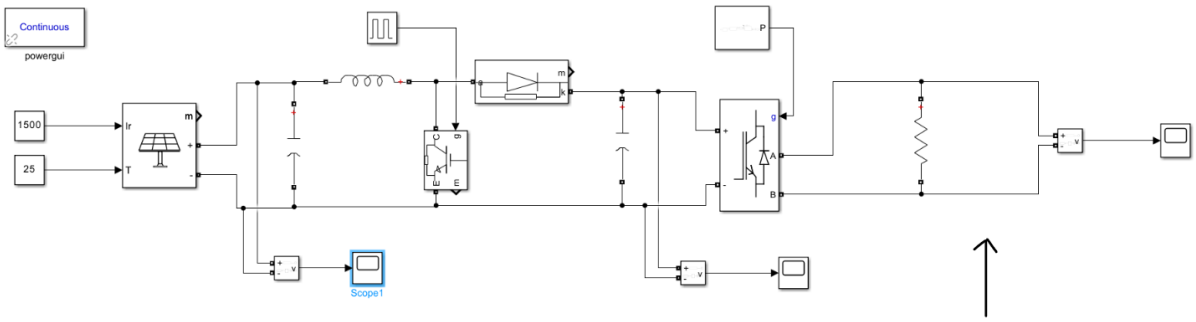
- DC link: record the capacitor value, measured voltage, and any voltage controller actually implemented.

- Inverter: record the bridge type, gate/PWM method, switching frequency, and measured output frequency.

- Output stage: the second configuration adds a passive output filter containing an inductor and capacitor. Record their values and the load resistance or other load parameters.

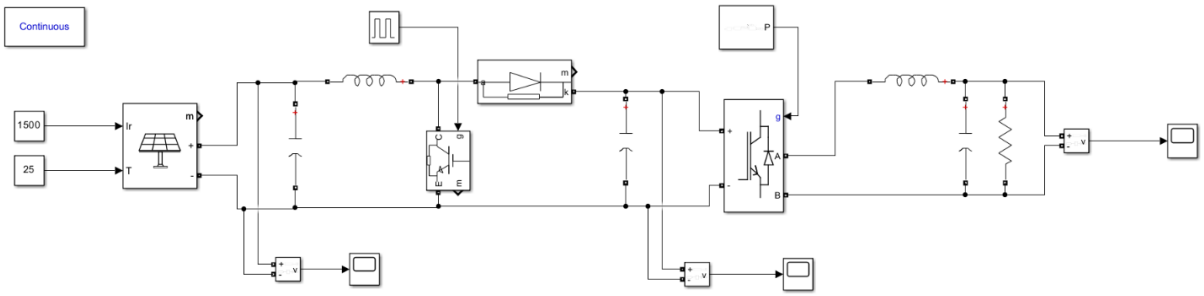
- Measurements: document the voltage-measurement locations and polarity, and identify which Scope corresponds to the pre-boost, post-boost, and output waveforms.

- **<u>Without Passive filter at the ouput side :</u>**



<u>Simulink Model</u>

- **<u>With Passive filter at the ouput side :</u>**



<u>Simulink Model</u>

5. Methodology :

1. Set the PV irradiance and temperature.
2. Configure electrical components and switching/control signals.
3. Set solver and simulation stop time appropriate to the switching frequency.
4. Run the baseline test.
5. Log PV, DC-link, inverter, and load measurements.
6. Repeat the simulation for irradiance, temperature, and load variations.
7. Calculate average power, ripple, and efficiency where suitable.
8. Compare results and explain deviations from ideal theory.

6. Results and discussion :

- **Without Passive filter at the ouput side :**

- **Graphs :**

  Configuration 1 — without output passive filter: The pre-boost voltage plot is nearly constant over the displayed time interval. The post-boost plot shows a dense band of switching-related variation. The output voltage plot contains repeated positive and negative switching pulses rather than a smooth sinusoidal waveform. This indicates that inverter switching components are visible at the load when no output passive filter is used. These plots are qualitative; numerical ripple and RMS values have not been provided.

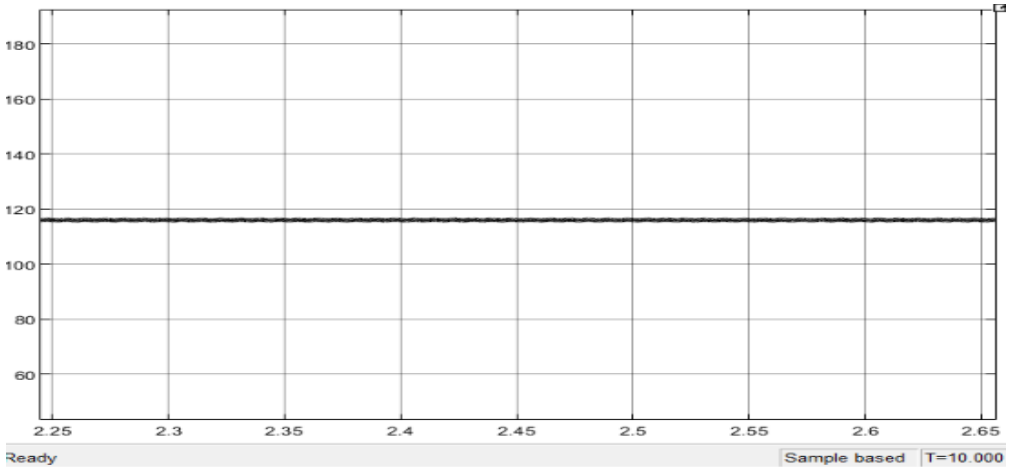


Fig . Voltage wavefrom before boosting

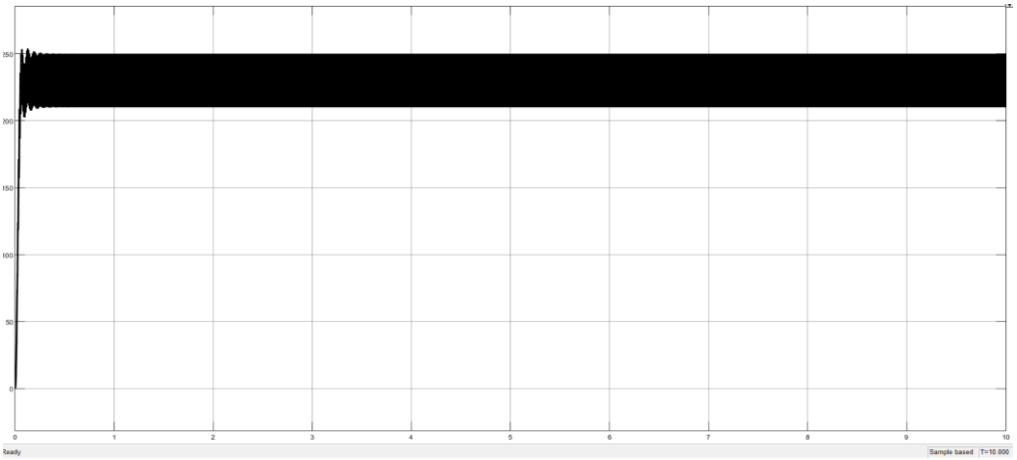


Fig . Voltage wavefrom after boosting

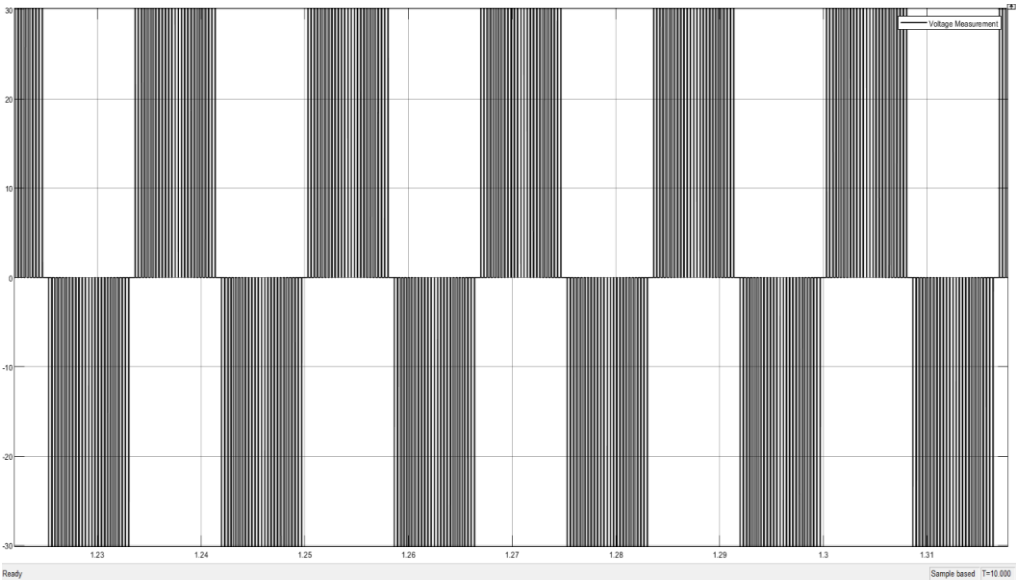
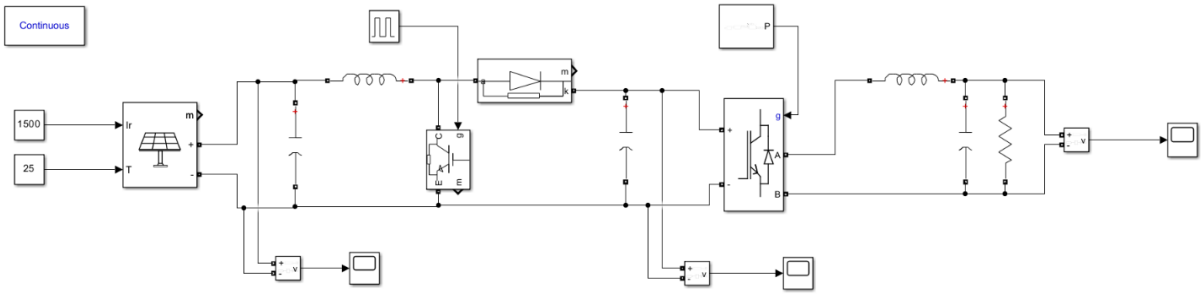
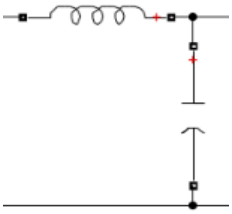


Fig . Output Voltage wavefrom

➢ With Passive filter at the ouput side :



Simulink Model



Passive filter

Configuration 2 — with output passive filter: The pre-boost voltage plot rises quickly and then remains nearly steady in the displayed interval. The post-boost voltage plot shows a startup transient that decays toward a steady value. The output voltage plot is approximately sinusoidal and smoother than the unfiltered output. This qualitative comparison is consistent with the passive filter attenuating high-frequency switching components. Exact filter effectiveness should be quantified using RMS measurements, ripple, or THD if those data are available.

- **Graphs :**

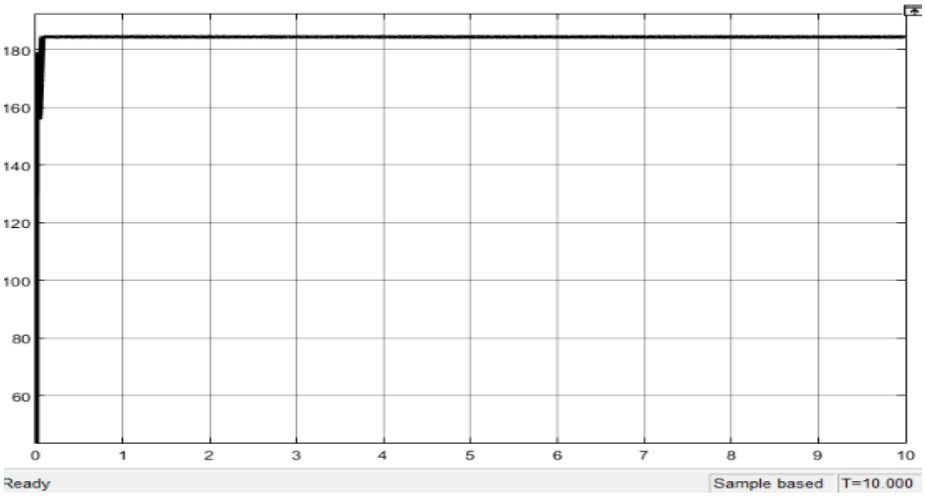


Fig . Voltage wavefrom before boosting

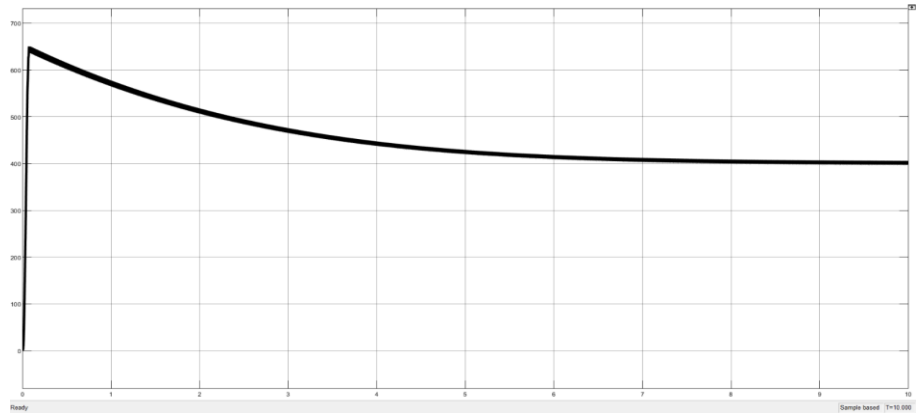


Fig . Voltage wavefrom after boosting

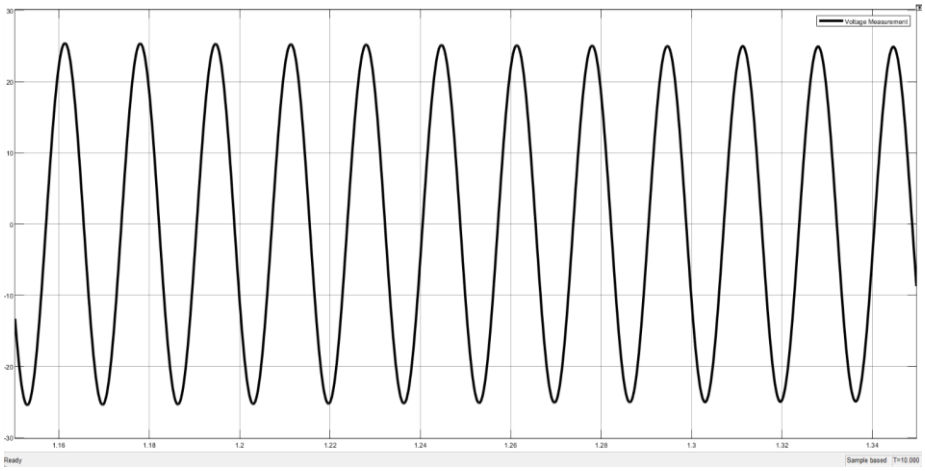


Fig . Output Voltage wavefrom

7. Limitations :

Comparison of the two output configurations: Without the passive filter, the inverter output displays prominent switching pulses. With the passive filter, the displayed output waveform is smoother and approximately sinusoidal. Therefore, the filter improves the visible output waveform in the supplied simulation plots. A quantitative claim about the amount of improvement cannot be made from the screenshots alone because the axis scales, RMS measurements, and THD values are not fully specified.

The current study focuses on the simulated PV, boost-converter, inverter, filter, and load arrangement. The plots supplied do not provide enough information to calculate conversion efficiency, total harmonic distortion (THD), voltage ripple, or settling time numerically. The model has not been documented as a grid-connected system. The control input shown in the model appears to be a pulse signal; unless an MPPT algorithm and feedback controller are present in the actual model, the system should not be described as using MPPT or closed-loop DC-link regulation. Device losses, parasitic elements, and detailed thermal effects should be discussed as limitations only if they are not included in the model.

8. Conclusion :

The MATLAB/Simulink model demonstrates the main stages of a solar PV power-conversion system: PV generation, DC–DC boost conversion, DC-link energy storage, DC-to-AC inversion, and delivery to a load. The supplied plots show the voltage before boosting, the voltage after boosting, and the inverter output. In the configuration without the output passive filter, the output waveform contains prominent switching pulses. With the passive filter, the output voltage appears smoother and approximately sinusoidal, indicating attenuation of high-frequency switching components. The plots support a qualitative comparison of

the two configurations, but numerical conclusions about efficiency, ripple, RMS voltage, or THD require measurements from exported simulation data.

9. Future scope :

- Implement and compare Perturb and Observe (P&O) or Incremental Conductance MPPT, if not already present in the current model.

- Add closed-loop DC-link voltage regulation and compare its transient response with the present control method.

- Compare converter efficiency across irradiance, temperature, and load conditions using measured average input and output power.

- Include realistic switching/device losses and parasitic elements to improve model fidelity.

- Evaluate output total harmonic distortion (THD) and compare passive-filter parameter choices.

- Extend the validated standalone system to include battery storage or a grid-connected inverter.

References :

[1] MathWorks, "PV Array," *Simulink Documentation*. Available: https://www.mathworks.com/help/sps/powersys/ref/pvarray.html

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